Split-Dependent Transfer in Multi-Target Bioactivity Prediction: A Leakage-Controlled ChEMBL 37 Benchmark and Locked Plant-Bag Extension

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Abstract

Background: Traditional-medicine databases record uses at plant level, whereas bioactivity databases measure compound–target relations. We establish a leakage-controlled activity benchmark and a prospectively locked multiple-instance learning (MIL) protocol for testing plant-level supervision without broadcasting a use label to every reported constituent.

Methods: We curated ChEMBL 37 measurements for human DPP4, acetylcholinesterase, and COX-2. A 0.70 Morgan-Tanimoto connected-component split defined the primary chemical out-of-distribution (OOD) test; a first-publication-year split and a continuous-pChEMBL endpoint were sensitivities. Logistic regression, Extra Trees, separate and shared fingerprint multilayer perceptrons (MLPs), and an official Chemprop D-MPNN were run with five fixed seeds. Selection and early stopping used validation data only; ranking, enrichment, discrimination, and calibration were computed on the sealed test set.

Results: Among 43,843 candidates, 16,941 records yielded 10,698 labelled pairs and 10,686 molecules after curation. No identity, Murcko scaffold, or chemical component crossed primary partitions; maximum test-to-train Tanimoto was 0.698. Extra Trees achieved the highest chemical-OOD macro PR-AUC (0.9425 ± 0.0009), followed by logistic regression (0.9334 ± 0.0000), separate MLPs (0.9281 ± 0.0035), shared MLP (0.9203 ± 0.0024), and D-MPNN (0.8938 ± 0.0063). The shared-minus-separate MLP difference was -0.0077 (cluster

bootstrap 95% CI -0.0141 to -0.0021), driven by acetylcholinesterase after multiplicity correction. Conversely, the shared MLP improved temporal macro PR-AUC (0.9425 ± 0.0024 versus 0.9247 ± 0.0029), but worsened continuous-pChEMBL MAE (0.8181 ± 0.0049 versus 0.8083 ± 0.0066).

Conclusions: Model ordering was endpoint-, split-, target-, and metric-dependent; simple fingerprint ensembles remained strong controls. The benchmark validates the activity branch and quantifies negative transfer, but does not test the ethnopharmacological MIL hypothesis, which remains locked pending redistributable VietHerb data and expert-audited therapeutic-use mappings.

Scientific Contribution: We provide a version-pinned, leakage-audited, five-seed benchmark that evaluates multi-target sharing under chemical-OOD, temporal, and continuous endpoints. Unlike naive herb-to-compound label propagation, the accompanying protocol formalizes plant knowledge as positive–unlabelled bag supervision and requires falsification controls before any ethnopharmacological claim.

Keywords: chemical out-of-distribution evaluation; ChEMBL; multi-task learning; negative transfer; multiple-instance learning; natural products; virtual screening

1 Introduction

Natural products remain an important source of chemical diversity for drug discovery, but their biological evidence is heterogeneous and frequently recorded at a different level from the molecule being screened. VietHerb was created to organize Vietnamese medicinal plants together with metabolites, therapeutic effects, morphology, and geography. Its original release reported 2,881 species, 10,887 metabolites, and many-to-many species–metabolite and species–effect relations [Nguyen-Vo et al., 2019]. This local resource is scientifically valuable, but a reported use of a plant is neither evidence that every constituent is active nor proof of clinical efficacy.

Prior computational ethnopharmacology already links traditional symptoms, efficacies, herbs, ingredients, and molecular targets. HTINet and HTINet2 use heterogeneous networks and knowledge graphs for herb–target prediction [Wang et al., 2019, Duan et al., 2024]; other systems use clinical symptom profiles or multimodal ingredient–target information [Cheung, 2026, Liang et al., 2025]. Consequently, the broad claim that traditional knowledge has not been used for target prediction is untenable. The unresolved issue addressed here is narrower: VietHerb annotations are plant-level labels, while ChEMBL observations are compound-level measurements. This is naturally a multiple-instance learning (MIL) problem, in which a plant is a bag and its reported

metabolites are instances [Ilse et al., 2018, Doran and Ray, 2012].

Reliable evaluation is equally important. Ligand benchmarks can reward memorization when close analogues cross partitions [Wallach and Heifets, 2018], and conventional scaffold splitting does not eliminate every form of chemical redundancy. Traditional-medicine data add botanical leakage because the same metabolite can occur in several species. Missing compound–target and plant–use edges also cannot automatically be interpreted as negatives, making positive–unlabelled (PU) learning a relevant sensitivity analysis [Kiryo et al., 2017, Wang et al., 2026].

We make three empirical and one prospective contribution. First, we release an auditable ChEMBL 37 cohort with connected-component chemical-OOD and temporal partitions. Second, we compare linear, tree-ensemble, fingerprint-neural, and message-passing baselines across ranking, early enrichment, discrimination, and calibration. Third, we test whether the sign of multi-target transfer persists under temporal classification and continuous potency prediction. Finally, we specify cross-granularity plant supervision as attention MIL with positive–unlabelled labels and explicit shuffled-label controls. This last hypothesis is deliberately not evaluated without a legally redistributable VietHerb snapshot and expert-audited mappings.

2 Materials and methods

2.1 Data sources and governance

VietHerb provides plant–metabolite and plant–therapeutic-effect relations [Nguyen-Vo et al., 2019]. Because no explicit database redistribution licence was found during our audit, the confirmatory analysis requires a maintainer-authorized export. Raw rows will not be redistributed without written permission. Published artifacts will otherwise contain code, schemas, source identifiers, checksums, aggregate counts, and split-generation logic. Traditional-use records are treated as hypothesis-generating cultural and historical evidence, not as validated biomedical claims.

For the confirmatory study, measured activities were extracted from the versioned ChEMBL 37 SQLite release; the archive, database, and output were bound by SHA-256 hashes. The earlier live REST snapshot is retained only as an exploratory engineering analysis. ChEMBL is a manually curated, large-scale bioactivity resource with explicit assay and document provenance [Zdrazil et al., 2024]. LOTUS may be used only for an optional occurrence/provenance sensitivity analysis [Rutz et al., 2022]. Aggregated natural-product databases are not assumed to provide independent activity ground truth [Chandrasekhar et al., 2025].

2.2 Target-activity cohort

The pilot targets are human dipeptidyl peptidase 4 (DPP4, CHEMBL284), acetylcholinesterase (AChE, CHEMBL220), and prostaglandin G/H synthase 2 (COX-2/PTGS2, CHEMBL230). The primary cohort contains high-confidence, single-protein biochemical binding assays with standardized IC_{50} or K_i values in nM. Functional assays form a separate sensitivity cohort. Records with data-validity warnings, duplicate flags, unresolved structures, or ambiguous target assignments are excluded. Exact standardized structure–target replicates are aggregated by median pChEMBL; pairs spanning at least one log unit are flagged as contradictory and excluded from the primary analysis.

The prespecified binary endpoint labels pChEMBL ≥ 6 as active and pChEMBL ≤ 5 as inactive; the interval $(5, 6)$ is excluded as a gray zone. Only exact (equality) measurements enter the primary binary cohort. Alternative cutoffs and continuous pActivity models are sensitivity analyses. Unmeasured compound–target pairs remain masked. All stereochemistry and structure standardization decisions, rejected rows, release metadata, and SHA-256 hashes are retained in a machine-readable manifest.

2.3 Therapeutic-use normalization

Original text is preserved. Candidate mappings are retrieved from frozen MeSH/Mondo releases, after which a language model may rank candidates but may not accept a label automatically. The stored record includes source span, ontology version, candidate list, confidence, model/prompt hash, abstention, and reviewer decisions. Two pharmacognosy or pharmacy reviewers independently annotate a prespecified stratified sample before adjudication. We report exact and ancestor-aware precision, recall, F1, raw agreement, and Cohen's κ; sample size is selected for a target confidence-interval width rather than fixed solely by convenience.

The first model uses three coarse, clinically interpretable families: metabolic/diabetes, cognition/neurological use, and pain/inflammation. ATC is not used as the primary disease ontology because it classifies active drug substances rather than diseases or symptoms.

2.4 Molecular and botanical split construction

Molecules are standardized globally and assigned once across all targets and plants. Two locked regimes answer different questions. The chemical-OOD regime groups molecules by a prespecified fingerprint clustering rule and reports train–test nearest-neighbour Tanimoto distributions. A

scaffold component split is retained as a reproducible secondary regime. The botanical cold-start regime holds out plant taxa and explicitly handles metabolites shared between taxa.

For the strict joint split, molecule nodes are joined when they share a chemical group or occur in the same plant. Entire transitive connected components are assigned to train, validation, or test. Thus, if compound A is chemically linked to B and B occurs with C in one species, all three remain in one partition. No feature transformation, loss weight, class prior, or hyperparameter is fitted using test data.

The primary activity benchmark joined standardized molecules when their radius-2 Morgan fingerprint Tanimoto similarity was at least 0.70. Whole connected components were allocated 70:15:15 with a fixed split seed. An audit then required zero overlap in standardized structure, InChIKey, Murcko scaffold, and connected-component identifier. This scaffold-disjoint property is stricter than the construction rule and was checked rather than assumed.

The temporal sensitivity used each molecule–target pair's earliest source document year. Training contained first reports through 2017, validation contained 2018–2021, and test contained 2022–2025. Measurements published after the relevant training or validation cutoff were censored before label aggregation. Unlike the primary regime, temporal splitting intentionally did not enforce scaffold disjointness: it asks how a model trained on historical chemistry behaves on later publications, including later measurements of known series.

2.5 Multiple-instance multi-task model

Let x_{ij} denote metabolite j reported in plant i, and let $h_{ij} = f_\theta(x_{ij})$ be its shared molecular representation. The primary heads predict measured compound–target activities with masked, class-weighted binary cross entropy. For each plant bag, gated attention computes

$$e_{ij} = w^\top \left[\tanh(V h_{ij}) \odot \sigma(U h_{ij})\right], \tag{1}$$

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})}, \tag{2}$$

$$z_i = \sum_j a_{ij} h_{ij}, \tag{3}$$

followed by one plant-use head per therapeutic family [Ilse et al., 2018]. Attention weights provide instance attribution but are not interpreted as causal explanations.

An observed plant use is positive and absence is unlabelled. Auxiliary heads therefore use non-negative PU risk [Kiryo et al., 2017]. For class prior π, positive risk R_p^+, positive-as-negative

risk R_p^-, and unlabelled negative risk R_u^-, the unbiased estimate is

$$R_{\mathrm{PU}} = \pi R_p^+ + R_u^- - \pi R_p^-, \tag{4}$$

with the non-negative correction applied when the empirical negative component falls below zero. Priors are frozen from training-only estimation and varied in sensitivity analyses. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{activity}} + \lambda \mathcal{L}_{\text{plant-MIL}}. \tag{5}$$

The compute-feasible encoder is a radius-2, 2,048-bit chiral Morgan fingerprint followed by a two-layer MLP. This deliberately separates the prospective MIL test from an expensive foundation model. The measured-activity benchmark also included an official Chemprop directed message-passing neural network (D-MPNN) [Yang et al., 2019] using identical split IDs.

2.6 Baselines and falsification controls

Measured-activity baselines were per-target L2-regularized logistic regression, per-target Extra Trees, separate target MLPs, a hard-sharing multi-target MLP, and a shared Chemprop 2.2.1 D-MPNN. Logistic regression and Extra Trees used the same Morgan features and split assignments as the MLPs. The D-MPNN consumed SMILES and three masked activity columns; missing compound–target observations did not contribute to its loss.

The future VietHerb matrix additionally contains naive plant-label broadcasting, bag-level MIL, shuffled-use-label MIL, prevalence-matched random auxiliary labels, and optional domain importance weighting. Controls shuffle labels only among training plants and equalize parameter and update budgets. The primary future comparison is MIL versus the architecture-matched activity-only model; naive broadcasting is an ablation, not the proposed method.

2.7 Training and statistical analysis

The MLP used hidden dimensions 1,024 and 512, dropout 0.25, AdamW with learning rate 3×10^{-4} and weight decay 10^{-4}, batch size 256, and early stopping patience 12 on validation macro PR-AUC. The D-MPNN used the official Chemprop 2.2.1 defaults, at most 100 epochs, patience 12, and validation loss for deterministic checkpoint selection. Validation PR-AUC was not used for the D-MPNN because exact-seed CUDA execution invokes a non-deterministic cumulative sum in that metric; all test metrics were instead recomputed after restoring the best validation-loss

checkpoint. Five fixed seeds were used: 13, 37, 71, 101, and 137.

Model and hyperparameter selection was validation-only. Each candidate was run on the validation partition across the prespecified seeds; the configuration with the best mean validation objective was frozen and only then evaluated on test. The selector rejects columns whose names indicate test outcomes. The future MIL auxiliary weights $\{0.1, 0.3, 1.0\}$ and PU priors follow this same policy.

The primary activity endpoint was macro test PR-AUC. We additionally reported target-wise PR-AUC and prevalence, BEDROC ($\alpha = 20$), EF1% and EF5%, precision/recall at fixed experimental budgets, AUROC, Brier score, ten-bin expected calibration error (ECE), and reliability curves. PR-AUC was emphasized for imbalanced screening [Saito and Rehmsmeier, 2015], while BEDROC measures early recognition [Truchon and Bayly, 2007]. Paired 95% intervals use 1,000 bootstrap replicates with structure cluster as the resampling unit for compound endpoints and plant as the unit for bag endpoints. Target-level secondary tests use Benjamini–Hochberg correction.

The continuous sensitivity retained gray-zone observations and predicted median pChEMBL. Mean absolute error (MAE) was the leading regression metric, supplemented by root mean squared error and Spearman correlation. The natural-product sensitivity joined the ChEMBL 37 molecule-dictionary flag generated from COCONUT 2025-07. Subgroup results were declared descriptive unless a target had at least 30 positives and 30 negatives.

Within each neural run, the test set was evaluated only after restoring the best validation checkpoint. The source, dataset, run, metric, and prediction artifacts are bound by SHA-256. Runs used Python 3.11.9, RDKit 2025.09.5, scikit-learn 1.8.0, PyTorch 2.5.1 with CUDA 12.1, and Chemprop 2.2.1 on an NVIDIA RTX 4060 Laptop GPU with 8 GB memory.

3 Results

3.1 Workflow and locked analysis boundaries

Figure 1 separates completed measured-activity experiments from the prospectively locked VietHerb arm. All numerical results below derive from ChEMBL 37 and carry `scientific_results=true` in their run manifests. No synthetic bag, VietHerb row, or live-API exploratory observation contributes to a reported estimate.

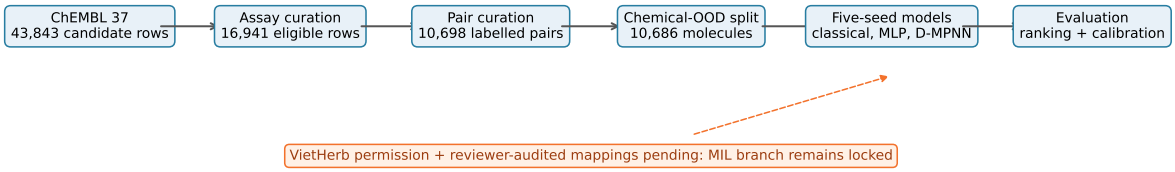


Figure 1: Study workflow and release gates. Completed ChEMBL analyses are distinguished from VietHerb-dependent experiments that remain prospectively locked.

3.2 Cohort and shift audit

The frozen ChEMBL 37 database yielded 43,843 candidate records for the three targets. Pre-specified assay, measurement, relation, validity, duplicate, and structure filters retained 16,941 rows. After standardization and replicate aggregation, 435 contradictory pairs and 3,353 gray-zone pairs were excluded, leaving 10,698 labelled structure–target pairs and 10,686 unique molecules. Chemical-component assignment produced 7,480 training, 1,603 validation, and 1,603 test molecules spanning 4,417 Murcko scaffolds. Pairwise overlap between every split was zero for molecule identifier, InChIKey, scaffold, and connected component. Test-to-train nearest-neighbour Tanimoto had median 0.500, 95th percentile 0.673, and maximum 0.698, below the locked 0.70 clustering threshold. Table 1 summarizes the target observations. No VietHerb row was used.

Table 1: Frozen ChEMBL 37 activity cohort after curation. Active denotes pChEMBL ≥ 6; inactive denotes pChEMBL ≤ 5. This table validates the confirmatory activity branch but contains no VietHerb supervision.

Endpoint	Measurements	Molecules	Active (%)	Test positives
DPP4	3,725	3,725	93.4	522
AChE	4,026	4,026	71.3	431
COX-2	2,947	2,947	77.8	344
All targets	10,698	10,686	80.8	1,297
Plant-use bags	0	0	—	0

The absence of identity, scaffold, and fingerprint-neighbour overlap satisfied all primary chemical leakage gates. The continuous cohort contained 14,029 molecules and 14,051 molecule–target pairs, including the binary gray zone, with 9,820/2,105/2,104 molecules in train/validation/test. The temporal cohort contained 10,697 molecules after excluding 60 rows without a document year and censoring 343 post-cutoff training or validation observations; its split was 7,896/1,734/1,067. Botanical bag sizes and plants per metabolite cannot be estimated until an authorized VietHerb

snapshot is frozen.

3.3 Primary chemical-OOD benchmark

Extra Trees gave the highest mean macro PR-AUC, followed by logistic regression, separate MLPs, the shared MLP, and D-MPNN (Table 2 and Figure 2). Thus neither hard parameter sharing nor graph message passing improved global ranking over the classical fingerprint baselines in this chemical-OOD regime. Model ordering differed for calibration: D-MPNN had the lowest macro Brier score (0.1396) and ECE (0.0525), whereas the shared MLP had the highest values (0.2253 and 0.2928; Figure 4). The D-MPNN therefore produced better-scaled probabilities despite lower PR-AUC.

Table 2: Frozen ChEMBL 37 chemical-OOD test performance, reported as mean ± standard deviation across five seeds. EF1% is an enrichment factor, not a percentage.

Model	Macro PR-AUC	BEDROC20	EF1%	Brier	ECE
Extra Trees	0.9425 ± 0.0009	0.9888	1.2527	0.1498	0.1943
Logistic regression	0.9334 ± 0.0000	0.9859	1.2527	0.1715	0.1593
Separate target MLPs	0.9281 ± 0.0035	0.9714	1.2334	0.1833	0.2140
Shared activity MLP	0.9203 ± 0.0024	0.9803	1.2527	0.2253	0.2928
Chemprop D-MPNN	0.8938 ± 0.0063	0.9311	1.1187	0.1396	0.0525

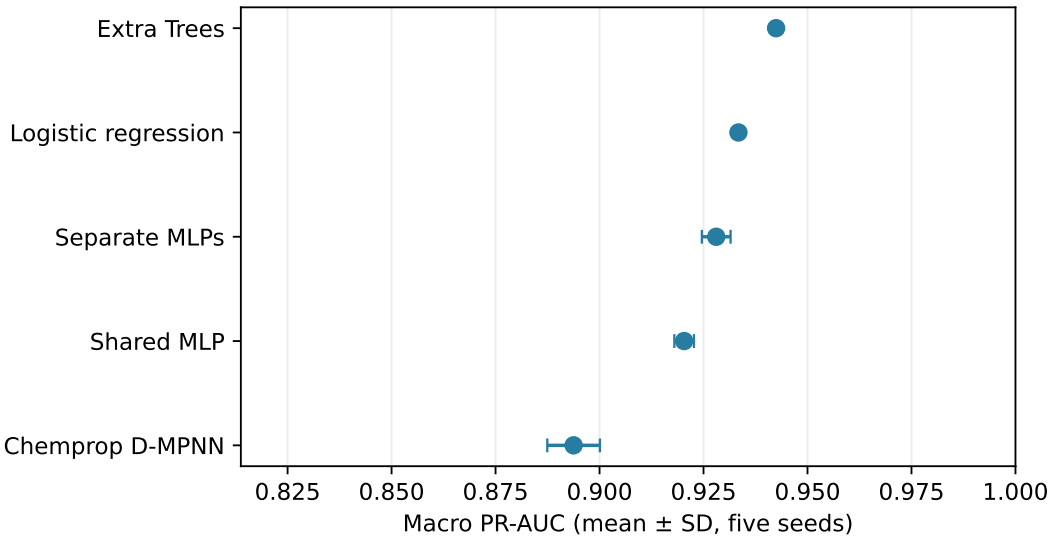


Figure 2: Chemical-OOD macro PR-AUC across five seeds. Points denote means and horizontal intervals show one standard deviation; the displayed axis spans 0.82–1.00.

3.4 Target-wise transfer and early recognition

For the architecture-matched neural comparison, separate target MLPs exceeded the shared activity-only MLP by 0.0077 macro PR-AUC. A hierarchical bootstrap over seed and chemical

component gave a 95% interval of −0.0141 to −0.0021 for shared minus separate. Target-wise differences were −0.0271 for AChE (95% CI −0.0463 to −0.0110, BH-adjusted $p < 0.001$), +0.0045 for DPP4 (95% CI −0.0015 to 0.0110, adjusted $p = 0.261$), and −0.0005 for PTGS2 (95% CI −0.0063 to 0.0050, adjusted $p = 0.900$). Negative transfer was therefore localized to AChE rather than uniform across targets (Figure 3).

The shared MLP nevertheless increased macro BEDROC20 from 0.9714 to 0.9803 and EF1% from 1.2334 to 1.2527. The earliest-ranked candidates could thus improve while global precision–recall and probability calibration deteriorated (Figure 5). These results make metric choice scientifically material rather than cosmetic.

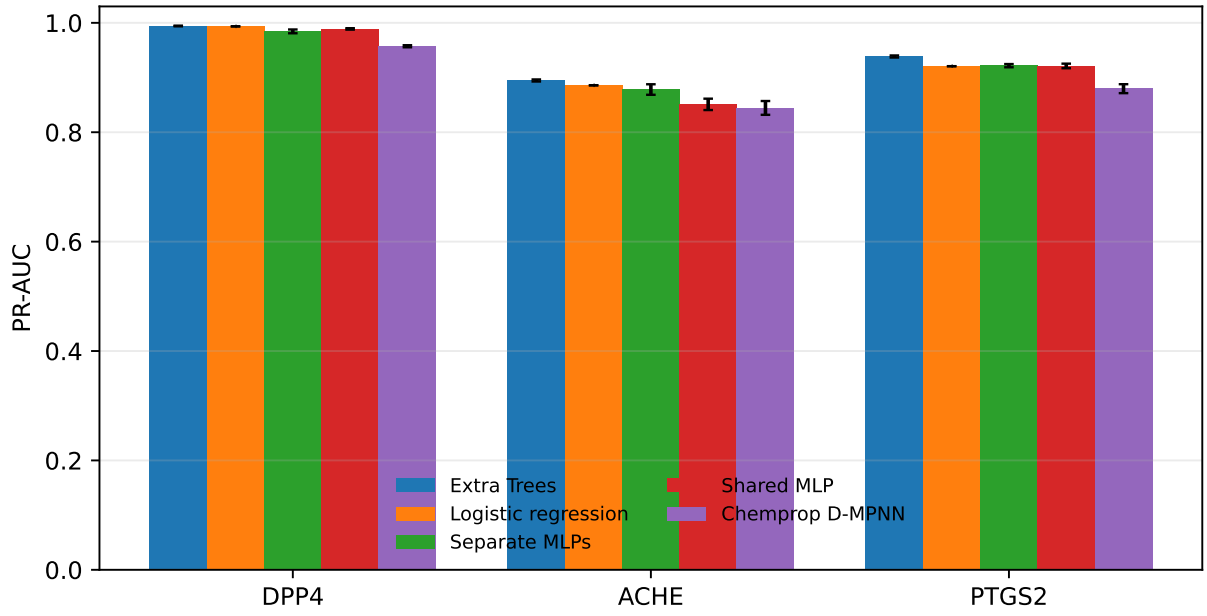


Figure 3: Target-level chemical-OOD PR-AUC. Error bars summarize five fixed seeds; AChE accounts for the corrected shared-model deficit.

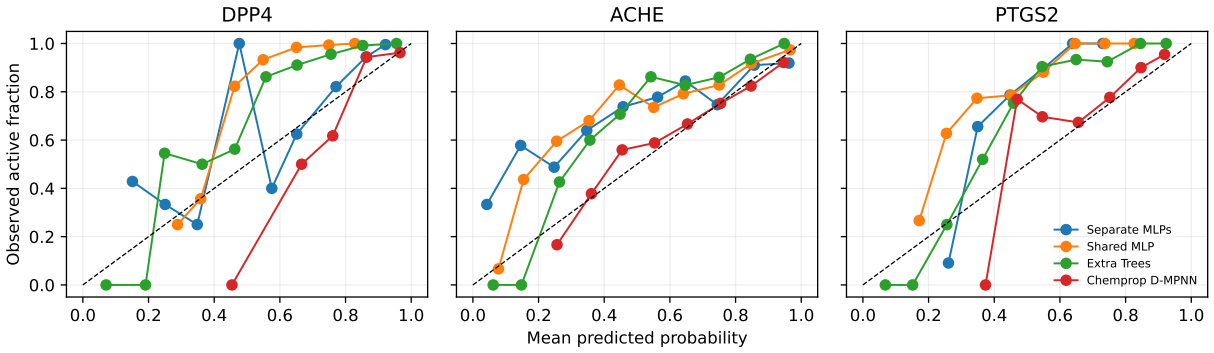


Figure 4: Reliability curves on the chemical-OOD test set. The D-MPNN was less accurate in ranking but substantially better calibrated.

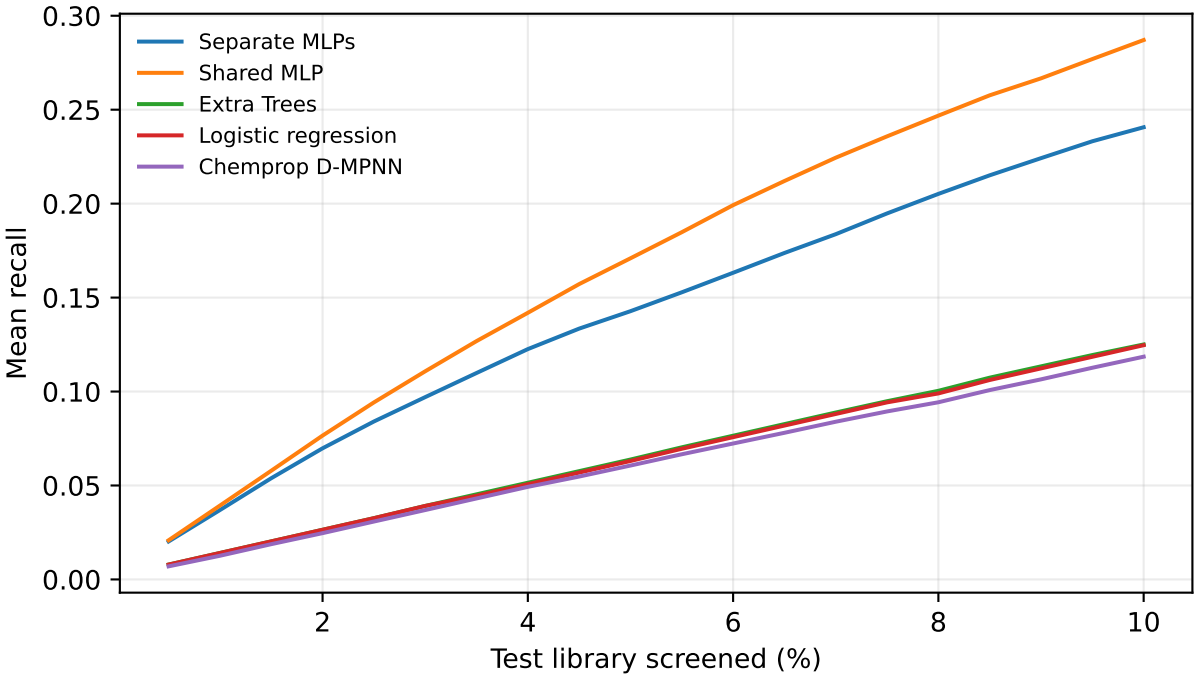


Figure 5: Recall as the screening budget increases. Curves average targets and seeds and complement the BEDROC20 and enrichment-factor summaries.

3.5 Temporal and continuous-endpoint sensitivity

The temporal experiment changed the conclusion about hard sharing. Extra Trees remained highest at 0.9432 ± 0.0008, but the shared MLP was nearly tied at 0.9425 ± 0.0024 and exceeded logistic regression (0.9257 ± 0.0000) and separate MLPs (0.9247 ± 0.0029). Shared-model target PR-AUC was 0.9035 for AChE, 0.9721 for DPP4, and 0.9518 for PTGS2, compared with 0.8993, 0.9792, and 0.8957 for separate models. The largest temporal sharing gain was consequently on PTGS2.

Continuous prediction restored the negative-transfer pattern. Across 14,051 pChEMBL observations, separate MLPs achieved macro MAE 0.8083 ± 0.0066, compared with 0.8181 ± 0.0049 for the shared model. Per-target mean MAE was 0.7990 versus 0.8153 for AChE, 0.7306 versus 0.7326 for DPP4, and 0.8954 versus 0.9064 for PTGS2. Figure 6 summarizes the two sensitivity analyses. Taken together, the transfer direction depended on both data split and endpoint definition.

3.6 Natural-product subgroup and external-validity boundary

The ChEMBL 37 molecule dictionary mapped 14,458 extracted molecules to a COCONUT 2025-07 natural-product flag; 657 were flagged overall and 402 occurred in the primary cohort. Only

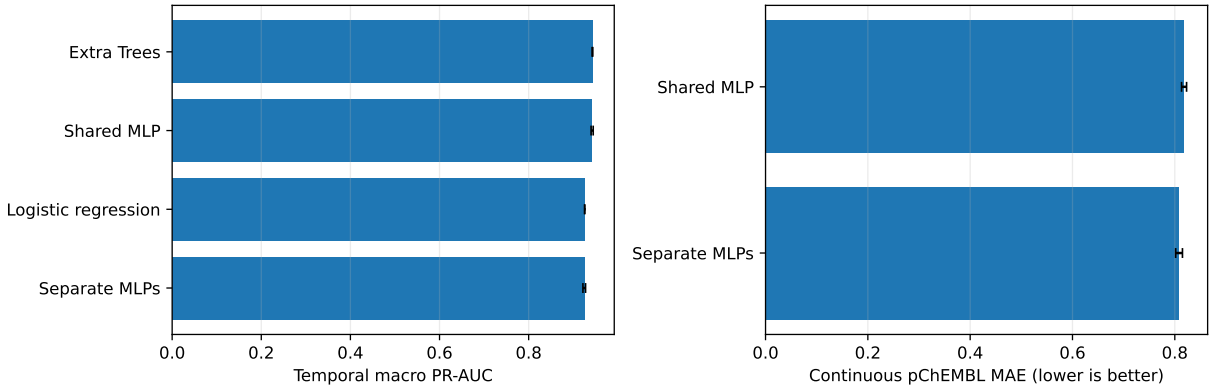


Figure 6: Temporal classification and continuous-potency sensitivity. Higher PR-AUC is better; lower MAE is better.

79 flagged molecules reached the primary test set. The target subsets comprised 55 AChE observations (11 active, 44 inactive), seven DPP4 observations (five active, two inactive), and 19 PTGS2 observations (nine active, ten inactive). None met the prespecified minimum of 30 positives and 30 negatives, so all subgroup scores are descriptive and no natural-product superiority claim is made.

We also assessed MoleculeACE [van Tilborg et al., 2022] as a possible external benchmark. Its 30 ChEMBL 29 activity-cliff datasets do not include CHEMBL220, CHEMBL230, or CHEMBL284 and originate from the same database family. It was therefore not relabelled as independent external validation. The temporal holdout measures chronological transport, but genuinely external matched-target biological validation remains open.

3.7 Implementation and confirmatory readiness

All 47 unit tests passed, including masked-loss, nnPU-risk, attention-MIL, metric, temporal split, validation-only selection, and connected-component leakage invariants. A CUDA smoke run exercised the activity-only, naive-broadcast, MIL, and shuffled-MIL paths end to end on deterministic synthetic bags. Its manifest is explicitly marked `scientific_results=false`; therefore synthetic performance is not reported. Botanical cold starts, PU-prior and auxiliary-weight sensitivity, and attention stability require real bags and remain locked confirmatory analyses. Detailed cohort audits, seed-level metrics, reliability bins, subgroup results, and artifact hashes are provided in Additional file 1 and accompanying machine-readable CSV files.

4 Discussion

This benchmark produced three practically important findings. First, the strongest chemical-OOD ranking model was Extra Trees on fixed Morgan fingerprints, not either neural architecture. This is a useful warning against treating representation complexity as a substitute for stringent splitting and strong classical controls. The D-MPNN result does not establish that molecular graphs are generally inferior; it shows that the tested official implementation and training budget did not overcome a well-tuned fixed representation on these three endpoints.

Second, performance was multidimensional. D-MPNN had the weakest macro PR-AUC but the best Brier score and ECE, while the shared MLP had competitive early enrichment despite worse global ranking and calibration. Screening programs that test only a small top-ranked fraction may reasonably prioritize BEDROC or EF, whereas calibrated probabilities matter for risk estimates and decision thresholds. Reporting only AUROC or a single aggregate score would have hidden these differences.

Third, multi-target transfer was not an invariant property of the targets. Hard sharing hurt chemical-OOD binary ranking and continuous potency prediction, yet helped under temporal classification. The primary deficit was concentrated in AChE; the temporal benefit was largest for PTGS2. A plausible interpretation is that shared features regularize chronological extrapolation within recurring chemical series but are less useful when all close fingerprint neighbours and scaffolds are removed. This is an inference from the observed split contrast, not a mechanistic conclusion, and should be tested across more targets.

The future VietHerb result must be interpreted as incremental predictive value, not validation of traditional medicine. If MIL improves over activity-only learning but not over shuffled labels, the gain is consistent with generic regularization rather than semantic ethnopharmacological signal. If naive broadcasting performs similarly to MIL, label granularity or leakage requires further investigation. If all ethnomedical variants underperform, negative transfer is an informative result and should not be hidden by selecting favorable targets. The present finding that even measured-activity sharing can reverse sign across regimes makes those controls essential.

Docking or short molecular dynamics may provide a later structure-based plausibility assessment, but neither constitutes biological validation. Docking scores have well-known limitations for affinity ranking [Warren et al., 2006]. Prospective biochemical assays with positive, negative, interference, and purity controls are required to claim target inhibition.

This study has several limitations. Three targets cannot establish broad cross-target generality,

and ChEMBL measurements retain inter-laboratory and assay heterogeneity after curation. The temporal holdout is not independent external validation, and the natural-product test subset was underpowered. The D-MPNN used one official version and budget rather than an exhaustive graph-model search. Bootstrap intervals quantify sampling variation under the chosen component structure but do not correct dataset-level selection bias. Finally, the central plant-bag experiment is not yet empirical: incomplete plant chemical inventories, uncertain traditional-use normalization, PU class-prior uncertainty, and non-causal attention remain open risks. Written data permission and domain review are scientific release gates, not administrative details.

5 Conclusion

We provide a version-pinned ChEMBL 37 benchmark in which 10,686 molecules passed identity, scaffold, and fingerprint-neighbour leakage audits. Extra Trees was the strongest chemical-OOD ranker, D-MPNN was the best-calibrated model, and the direction of multi-target transfer changed between chemical-OOD, temporal, and continuous endpoints. These findings argue for split-aware, target-wise, and multi-metric evaluation with classical baselines. We also provide a tested plant-bag MIL implementation and locked analysis plan, but the claim that ethnopharmacological supervision improves molecular prioritization remains neither supported nor refuted pending redistributable VietHerb data and expert-audited mappings.

Additional files

Additional file 1 (.pdf): supplementary methods, cohort audits, seed-level results, target-wise intervals, reliability summaries, natural-product subgroup counts, computational environment, and artifact provenance. Machine-readable tables (.csv) and a SHA-256 evidence manifest accompany the archived release.

List of abbreviations

AChE, acetylcholinesterase; BH, Benjamini–Hochberg; D-MPNN, directed message-passing neural network; DPP4, dipeptidyl peptidase 4; ECE, expected calibration error; EF, enrichment factor; MAE, mean absolute error; MIL, multiple-instance learning; MLP, multilayer perceptron; OOD, out-of-distribution; PR-AUC, area under the precision–recall curve; PTGS2, prostaglandin G/H synthase 2; PU, positive–unlabelled.

Declarations

Availability of data and materials

Code, configurations, schemas, tests, source locks, and derived manifests are available in the project repository: `https://github.com/MinhNhat-2504/vietherb-mtl`. ChEMBL-derived artifacts follow CC BY-SA 3.0 and can be regenerated from the public ChEMBL 37 SQLite release (`https://doi.org/10.6019/CHEMBL.database.37`). A versioned software and evidence archive will be cited by Zenodo DOI before submission. No VietHerb row supports the conclusions reported here. VietHerb row-level data will be used in a later study only if its maintainers authorize a reproducible access route.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Nhat Trinh Ngoc Minh: conceptualization, methodology, software, investigation, data curation, validation, visualization, and writing—original draft. Linh Nguyen Thi My: supervision and writing—review and editing. Both authors read and approved the final manuscript.

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Not applicable.

Ethics approval and consent to participate

The reported work used public chemical bioactivity records and deterministic synthetic bags for software validation; it involved no human participants, animals, or clinical intervention. Traditional-use annotations are not treated as clinical evidence.

Consent for publication

Not applicable.

Generative-AI assistance disclosure

A generative-AI coding assistant was used for implementation support, test generation, literature-search assistance, and language editing. The authors reviewed the source code, citations, numerical outputs, and manuscript claims and remain responsible for the final work.

References

Venkata Chandrasekhar, Kohulan Rajan, Sri Ram Sagar Kanakam, et al. COCONUT 2.0: a comprehensive overhaul and curation of the collection of open natural products database. *Nucleic Acids Research*, 53(D1):D634–D643, 2025. doi: 10.1093/nar/gkae1063.

Y. Cheung. NovelHTI: An interpretable pathway-enhanced framework for de novo target prediction of medicinal herbs. *Pharmaceuticals*, 19:413, 2026. doi: 10.3390/ph19030413.

Gary Doran and Soumya Ray. Multiple-instance learning from distributions. *BMC Bioinformatics*, 13:S15, 2012. doi: 10.1186/1471-2105-13-S15-S3.

Peng Duan et al. HTINet2: herb-target prediction via knowledge graph embedding and residual-like graph neural network. *Briefings in Bioinformatics*, 25:bbae414, 2024. doi: 10.1093/bib/bbae414.

Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80, pages 2127–2136. PMLR, 2018.

Ryuichi Kiryo, Gang Niu, Marthinus C. du Plessis, and Masashi Sugiyama. Positive-unlabeled learning with non-negative risk estimator. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

X. Liang et al. Herbal ingredient-target interaction prediction via multi-modal learning. *Information Sciences*, 711:122115, 2025. doi: 10.1016/j.ins.2025.122115.

Thanh-Hoang Nguyen-Vo, Tri Le, Duy Pham, et al. VIETHERB: A database for vietnamese herbal species. *Journal of Chemical Information and Modeling*, 59(1):1–9, 2019. doi: 10.1021/acs.jcim.8b00399.

Adriano Rutz, Maria Sorokina, Jakub Galgonek, et al. The LOTUS initiative for open knowledge management in natural products research. *eLife*, 11:e70780, 2022. doi: 10.7554/eLife.70780.

Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10:e0118432, 2015. doi: 10.1371/journal.pone.0118432.

Jean-Francois Truchon and Christopher I. Bayly. Evaluating virtual screening methods: Good and bad metrics for the early recognition problem. *Journal of Chemical Information and Modeling*, 47:488–508, 2007. doi: 10.1021/ci600426e.

Derek van Tilborg, Anna Alenicheva, and Francesca Grisoni. MoleculeACE: a benchmark for molecular activity cliff estimation. *Journal of Chemical Information and Modeling*, 62(20): 4891–4901, 2022. doi: 10.1021/acs.jcim.2c01073.

Izhar Wallach and Abraham Heifets. Most ligand-based classification benchmarks reward memorization rather than generalization. *Journal of Chemical Information and Modeling*, 58:916–932, 2018. doi: 10.1021/acs.jcim.7b00403.

Neng Wang et al. Herb target prediction based on representation learning of symptom related heterogeneous network. *Computational and Structural Biotechnology Journal*, 17:282–290, 2019. doi: 10.1016/j.csbj.2019.02.002.

Y. Wang et al. ProDTI: Prior-guided positive-unlabeled learning for robust drug-target interaction prediction. *Bioinformatics*, 42(10):btag712, 2026. doi: 10.1093/bioinformatics/btag712.

Gregory L. Warren, C. Webster Andrews, Anna-Maria Capelli, et al. A critical assessment of docking programs and scoring functions. *Journal of Medicinal Chemistry*, 49:5912–5931, 2006. doi: 10.1021/jm050362n.

Kevin Yang, Kyle Swanson, Wengong Jin, et al. Analyzing learned molecular representations for property prediction. *Journal of Chemical Information and Modeling*, 59:3370–3388, 2019. doi: 10.1021/acs.jcim.9b00237.

Barbara Zdrazil, Eloy Felix, Fiona Hunter, et al. The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52 (D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.